

LAUGHLIN BULLHEAD, AZ, USA

WMO: 723788

Lat: 35.157N

Lon: 114.559W

Elev: 212

StdP: 98.81

Time Zone: -7.00 (W07)

Period: 97-19

WBAN: 53135

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	2.9	4.9	-17.4	0.8	16.4	-15.0	1.0	17.2	16.2	11.0	15.1	12.0	2.6	30	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	45.0	20.3	43.7	20.1	42.5	19.9	23.6	35.0	23.0	35.1	22.5	35.4	4.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	16.0	27.3	19.1	14.2	28.9	17.7	13.1	30.7	71.1	34.3	68.7	34.5	66.8	36.1	34.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	12.0	10.9		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.7	13.3	15.2	19.1	22.5	26.7	32.9	35.5	34.7	31.0	24.1	17.3	12.3
	DBStd	8.78	3.22	3.62	3.56	3.74	3.87	3.34	2.43	2.74	3.13	3.73	3.68	3.39
	HDD10.0	30	10	3	0	0	0	0	0	0	0	0	1	16
	HDD18.3	556	158	98	36	9	1	0	0	0	0	3	62	189
	CDD10.0	5049	112	147	281	374	518	687	791	767	629	438	219	86
	CDD18.3	2532	1	9	58	133	259	437	532	508	379	183	31	1
	CDH23.3	39379	3	97	655	1613	3608	7446	9368	8705	5670	1954	258	3
	CDH26.7	26114	0	12	225	748	2068	5222	6917	6305	3647	909	61	0
Wind	WSAvg	4.1	5.5	4.4	4.1	4.2	4.4	3.9	3.6	3.2	3.1	3.9	4.2	5.0
Precipitation	PrecAvg	127	20	25	14	4	1	1	5	7	7	8	7	11
	PrecMax	280	108	119	111	18	14	9	35	28	41	56	32	49
	PrecMin	20	0	0	0	1	0	0	0	0	0	0	0	0
	PrecStd	74	30	29	25	4	3	2	8	8	10	15	9	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	28.0	33.8	37.5	42.0	47.7	46.4	45.2	42.8	37.6	30.2	23.0
		MCWB	10.7	12.6	14.9	15.8	17.6	20.1	20.7	21.0	20.0	18.1	14.5	12.6
	2%	DB	22.2	26.2	31.3	35.0	39.8	44.8	45.0	44.0	41.2	35.0	28.7	22.0
		MCWB	10.1	11.9	14.1	14.9	16.8	18.9	20.8	20.7	20.1	16.7	14.1	11.4
	5%	DB	20.1	24.1	29.0	32.8	37.7	42.8	43.6	42.8	39.9	33.0	27.2	19.7
		MCWB	9.4	11.3	13.0	14.1	16.0	18.9	20.7	20.4	19.6	16.0	13.1	10.0
	10%	DB	18.9	22.3	27.4	31.2	35.9	41.9	42.4	42.2	38.7	32.1	24.1	17.9
		MCWB	9.1	10.5	12.3	13.6	15.4	18.3	20.3	20.3	19.2	16.2	11.9	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	15.3	15.4	16.8	18.0	22.3	23.8	24.4	24.4	20.1	15.6	14.8
		MCDB	17.5	23.0	30.1	33.8	37.7	39.9	35.2	35.2	32.7	29.1	26.4	19.5
	2%	WB	12.2	13.4	14.5	15.7	17.0	21.0	23.2	23.8	23.2	19.0	14.6	13.2
		MCDB	17.5	21.2	28.2	31.2	35.9	40.5	35.7	35.5	32.3	29.4	26.3	18.3
	5%	WB	11.4	12.2	13.8	15.0	16.4	20.0	22.8	23.2	22.3	18.2	13.6	11.9
		MCDB	17.3	21.5	27.0	29.9	34.6	40.5	36.1	35.2	33.0	28.9	25.3	17.6
	10%	WB	10.2	11.3	12.8	14.1	15.7	19.0	22.4	22.7	21.6	17.2	12.7	10.4
		MCDB	16.5	21.1	26.0	29.4	33.4	39.8	36.5	35.3	33.3	28.1	23.9	15.9
Mean Daily Temperature Range	5% DB	MDBR	9.3	11.0	12.9	13.3	13.1	14.2	12.5	12.8	13.1	12.0	11.1	8.8
		MCDBR	11.4	14.4	16.3	16.4	16.1	15.7	13.8	14.0	14.8	13.8	13.7	11.0
		MCWBR	5.4	6.4	6.5	6.4	5.8	5.2	4.4	4.6	5.0	5.2	5.9	5.2
	5% WB	MCDBR	9.0	12.8	14.9	14.5	14.7	14.2	10.9	11.1	11.4	11.2	12.3	9.2
		MCWBR	4.5	5.8	6.0	5.5	5.5	4.7	3.3	3.7	4.5	4.0	5.6	4.5
Clear-Sky Solar Irradiance	taub		0.279	0.284	0.298	0.337	0.349	0.348	0.399	0.396	0.350	0.311	0.284	0.278
	taud		2.579	2.583	2.523	2.390	2.387	2.392	2.362	2.364	2.457	2.542	2.586	2.593
	Ebn at Noon		940	972	979	947	933	929	880	878	909	925	926	920
	Edh at Noon		76	85	98	118	120	120	123	120	104	87	75	71
All-Sky Solar Radiation	RadAvg		3.22	4.19	5.68	7.08	8.03	8.57	7.55	7.00	6.22	4.76	3.59	2.86
	RadStd		0.23	0.33	0.26	0.27	0.30	0.29	0.36	0.32	0.24	0.24	0.15	0.20

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	-1.71	N/A	-0.40	N/A	N/A	N/A	+166	+78

Nomenclature: See separate page